

Submission C solution to Problem 3

Abstract

We determine all $p \in [0, 1]$ such that, for every finite probability vector $(w_1, \dots, w_m)$ and i.i.d. Bernoulli(p) random variables $v_1, \dots, v_m$, one has

$$\mathbb{P}\left(\sum_{i=1}^m w_i v_i \geq p\right) \geq p.$$

The answer is

$$p \in [0, 1/3] \cup \{1/2, 1\}.$$

External inequalities used

We use two standard sharp inequalities.

Lemma 1 (Hoeffding–Pinelis comparison for weighted Bernoulli sums). *Let $0 < \theta < 1$, let $\xi_1, \dots, \xi_m$ be i.i.d. Bernoulli(θ), and let $a_i \geq 0$, $\sum_i a_i = 1$. Then for every real x ,*

$$\mathbb{P}\left(\sum_{i=1}^m a_i \xi_i < x\right) \leq \sup_{n \geq 1} \mathbb{P}\left(\frac{1}{n} B_{n, \theta} < x\right),$$

where $B_{n, \theta} \sim \text{Bin}(n, \theta)$.

This is the Bernoulli specialization of Hoeffding’s extremal theorem for sums of bounded independent random variables; see Hoeffding [1, Theorem 4]. For the left-tail formulation in essentially the above form, see also Pinelis [3, Corollary 3.7].

Lemma 2 (Ramanujan–Jogdeo–Samuels binomial inequality). *If $0 \leq \theta \leq 1/3$ and $B_{n, \theta} \sim \text{Bin}(n, \theta)$, then*

$$\mathbb{P}(B_{n, \theta} < n\theta) \leq 1 - \theta$$

for every integer $n \geq 1$.

Equivalently,

$$\mathbb{P}(B_{n, \theta} \geq n\theta) \geq \theta.$$

This is a classical binomial-tail inequality proved by Jogdeo and Samuels [2]; it is often discussed as a Ramanujan-type binomial inequality.

Main theorem

Theorem 1. *Let $p \in [0, 1]$. The following assertion holds for every $m \geq 1$, every choice of non-negative weights $w_1, \dots, w_m$ with $\sum_i w_i = 1$, and i.i.d. Bernoulli(p) random variables $v_1, \dots, v_m$:*

$$\mathbb{P}\left(\sum_{i=1}^m w_i v_i \geq p\right) \geq p.$$

This is true if and only if

$$p \in [0, 1/3] \cup \{1/2, 1\}.$$

Proof. We prove sufficiency and necessity separately.

Sufficiency for $0 \leq p \leq 1/3$. The case $p = 0$ is trivial, since the event $\sum_i w_i v_i \geq 0$ has probability 1. Assume $0 < p \leq 1/3$, and set

$$S = \sum_{i=1}^m w_i v_i.$$

By Lemma 1, with $\theta = p$ and $x = p$,

$$\mathbb{P}(S < p) \leq \sup_{n \geq 1} \mathbb{P}(B_{n,p} < np).$$

By Lemma 2, each term in the supremum is at most $1 - p$. Therefore

$$\mathbb{P}(S < p) \leq 1 - p,$$

and hence

$$\mathbb{P}(S \geq p) = 1 - \mathbb{P}(S < p) \geq p.$$

Sufficiency for $p = 1/2$. Let

$$S = \sum_{i=1}^m w_i v_i.$$

Since $v_i \sim \text{Bernoulli}(1/2)$, the vector $(1 - v_1, \dots, 1 - v_m)$ has the same distribution as $(v_1, \dots, v_m)$. Thus $1 - S$ has the same distribution as S . Hence

$$\mathbb{P}(S > 1/2) = \mathbb{P}(S < 1/2).$$

Consequently,

$$\mathbb{P}(S \geq 1/2) = \mathbb{P}(S > 1/2) + \mathbb{P}(S = 1/2) = \frac{1 + \mathbb{P}(S = 1/2)}{2} \geq \frac{1}{2}.$$

Sufficiency for $p = 1$. If $p = 1$, then $v_i = 1$ almost surely for every i , so

$$\sum_i w_i v_i = \sum_i w_i = 1$$

almost surely. Thus

$$\mathbb{P}\left(\sum_i w_i v_i \geq 1\right) = 1.$$

Necessity: failure for $1/3 < p < 1/2$. Take $m = 3$ and $w_1 = w_2 = w_3 = 1/3$. Let

$$K = v_1 + v_2 + v_3 \sim \text{Bin}(3, p).$$

For $1/3 < p < 1/2$,

$$\frac{K}{3} \geq p \iff K \geq 2.$$

Therefore

$$\mathbb{P}\left(\frac{K}{3} \geq p\right) = \mathbb{P}(K \geq 2) = 3p^2(1-p) + p^3 = p^2(3-2p).$$

Now

$$p^2(3-2p) < p$$

is equivalent to

$$p(3-2p) < 1,$$

or

$$2p^2 - 3p + 1 > 0,$$

i.e.

$$(2p-1)(p-1) > 0.$$

This holds for $1/3 < p < 1/2$. Hence the desired inequality fails throughout $(1/3, 1/2)$.

Necessity: failure for $1/2 < p < 1$. Take $m = 2$ and $w_1 = w_2 = 1/2$. Let

$$K = v_1 + v_2 \sim \text{Bin}(2, p).$$

For $p > 1/2$,

$$\frac{K}{2} \geq p \iff K = 2.$$

Thus

$$\mathbb{P}\left(\frac{K}{2} \geq p\right) = p^2 < p$$

for every $p \in (1/2, 1)$. Hence the desired inequality fails throughout $(1/2, 1)$.

Combining the sufficiency and necessity parts proves the theorem. $\square$

References

- [1] W. Hoeffding, On the distribution of the number of successes in independent trials, *Annals of Mathematical Statistics* **27** (1956), 713–721.
- [2] K. Jogdeo and S. M. Samuels, Monotone convergence of binomial probabilities and a generalization of Ramanujan’s equation, *Annals of Mathematical Statistics* **39** (1968), 1191–1195.
- [3] I. Pinelis, Optimal binomial, Poisson, and normal left-tail domination for sums of nonnegative random variables, *Electronic Journal of Probability* **21** (2016), article no. 47.